manipulation would actually be less successful in passing on genes, because of the economic costs of resisting.

But we have once again slipped back into looking at life from the point of view of the individual organism rather than its genes. When we talked about flukes and snails we accustomed ourselves to the idea that a parasite's genes could have phenotypic effects on the host's body, in exactly the same way as any animal's genes have phenotypic effects on its 'own' body. We showed that the very idea of an 'own' body was a loaded assumption. In one sense, all the genes in a body are 'parasitic' genes, whether we like to call them the body's 'own' genes or not. Cuckoos came into the discussion as an example of parasites not living inside the bodies of their hosts. They manipulate their hosts in much the same way as internal parasites do, and the manipulation, as we have now seen, can be as powerful and irresistible as any internal drug or hormone. As in the case of internal parasites, we should now rephrase the whole matter in terms of genes and extended phenotypes.

In the evolutionary arms race between cuckoos and hosts, advances on each side took the form of genetic mutations arising and being favoured by natural selection. Whatever it is about the cuckoo's gape that acts like a drug on the host's nervous system, it must have originated as a genetic mutation. This mutation worked via its effect on, say, the colour and shape of the young cuckoo's gape. But even this was not its most immediate effect. Its most immediate effect was upon unseen chemical happenings inside cells. The effect of genes on colour and shape of gape is itself indirect. And now here is the point. Only a little more indirect is the effect of the same cuckoo genes on the behaviour of the besotted host. In exactly the same sense as we may speak of cuckoo genes having (phenotypic) effects on the colour and shape of cuckoo gapes, so we may speak of cuckoo genes having (extended phenotypic) effects on host behaviour. Parasite genes can have effects on host bodies, not just when the parasite lives inside the host where it can manipulate by direct chemical means, but when the parasite is quite separate from the host and manipulates it from a distance. Indeed, as we are about to see, even chemical influences can act outside the body.

Cuckoos are remarkable and instructive creatures. But almost any wonder among the vertebrates can be surpassed by the insects. They have the advantage that there are just so many of them; my colleague Robert May has aptly observed that 'to a good approximation, all species